

Physics Guided Latent Dehazing Framework with Adaptive Gating and Frequency-Aware Refinement

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0.1 Implementation details

The proposed model was trained on 256×256 size random patches for synthetic haze training. A batch size of 16 was used, and the model was trained for 220 epochs. AdamW was employed as the optimizer, featuring an initial learning rate of 1×10^{-4} with a minimum learning rate of 5×10^{-6} and a weight decay of 1×10^{-4} . For imparting stability and reliable model selection, an exponential moving average (EMA) with decay of 0.999 and gradient clipping with a maximum norm of 0.8 were employed during training. The synthetic training phase employed four deterministic latent refinement steps utilizing $\beta = 0.10$ for regulated residual updates, residual clipping of 0.22 for stability, and a gate floor coefficient of 0.75 for allowing the adaptive gate to adjust the refinement strength while preserving critical residual corrections.

For real haze fine-tuning setup, a multi-scale cropping strategy was employed to enhance the robustness with limited training data. Hazy and clear images were randomly cropped at one of the three spatial resolutions (256×256 , 384×384 , 512×512). Training used the AdamW optimizer with a batch size of 1, two-step gradient accumulation, a starting learning rate of 5×10^{-5} , a minimum learning rate of 1×10^{-4} , a weight decay of 1×10^{-4} and 320 epochs were used for training with early stopping (patience 20). Here, 4-step latent refinement was used with β ranging from 0.08 to 0.12. Latent residual clipping of 0.2 was used. The proposed model is implemented

using PyTorch and trained on a Google Colab Linux virtual machine equipped with an NVIDIA A100-SXM4-80GB GPU.

0.2 Ablation Studies

To evaluate the effectiveness of various components of the proposed dehazing framework, an ablation study was conducted on the NH-Haze [1] dataset by removing one module at a time while preserving the remaining architecture and training parameters. The components evaluated include conditional latent refinement, wavelet HF injection with wavelet loss, dark channel guidance with dark channel loss, and the adaptive gate.

With a PSNR of 19.51 dB and SSIM of 0.665, Table 1 shows that the comprehensive model achieves optimal performance in haze removal, highlighting the collaborative function of these components in mitigating dense and no-uniform haze.

The removal of the latent refinement stage exhibits a performance reduction of 18.911 dB PSNR and 0.660 SSIM, indicating that despite the global structure is preserved by the coarse dehazing backbone, it is unable to address residual haze, color bias, and local restoration inaccuracies. Thus, subsequent to the coarse dehazing process, the latent refinement stage largely improves fine residual correction, which is in line with latent and efficient diffusion-based restoration frameworks [2], [3].

The PSNR and SSIM reduces to 19.067 dB and 0.632, respectively, without wavelet HF

Table 1: An ablation study on NH-Haze analyzing the contribution in dehazing performance of various components such as adaptive gated mechanism, wavelet high frequency (HF) guidance, latent refinement, and dark channel guidance evaluated by PSNR (dB) and SSIM.

Variant	Latent refinement	Wavelet HF injection + wavelet loss	Dark channel + dark channel loss	Adaptive gate	PSNR	SSIM
w/o adaptive gate	✓	✓	✓	×	19.095	0.629
w/o wavelet HF	✓	×	✓	✓	19.067	0.632
w/o dark channel	✓	✓	×	✓	18.732	0.631
w/o latent refinement	×	✓	✓	×	18.911	0.660
Full model	✓	✓	✓	✓	19.510	0.665

Table 2: Analysis of the 4-step and 6-step PLAGF-Dehaze configurations on both real and synthetic haze datasets demonstrating the influence of latent refinement depth

Method	NH-Haze		O-Haze		SOTS Indoor		SOTS Outdoor		Haze4K		Parameters (M)
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	–
Proposed (4-step)	19.20	0.630	22.33	0.733	27.00	0.950	28.00	0.958	23.32	0.930	3.52
Proposed (6-step)	19.51	0.665	22.30	0.732	27.01	0.953	27.90	0.958	23.72	0.930	3.52

injection and wavelet HF loss, signifying that frequency domain guidance contributes in preserving edges, textures, and structural information, thereby reducing over-smoothing. It helps in recovering high-frequency components diminished by haze. This observation aligns with spatial frequency dehazing research, which utilizes DWT-based decomposition, especially for non-uniform haze removal [4].

A performance reduction with 18.732 dB PSNR and 0.631 SSIM is witnessed upon removal of dark channel guidance and dark channel loss from both the coarse dehazing and latent refinement stages. This demonstrates the effectiveness of the valuable haze-density awareness provided by dark channel guidance to both coarse dehazing and latent refinement stages, especially for non-uniform haze removal. This is consistent with prior driven diffusion dehazing methods, where physical priors improve stability and efficient haze removal [5].

The removal of the adaptive gate reduces performance with 19.095 dB PSNR and 0.629 SSIM, showing the significance of spatially regulated residual refinement for dense, non-uniform

hazy scenes where different regions require varying levels of refinement. Without the adaptive gate which is derived from latent conditioning features the latent residual is refined more uniformly, causing insufficient refinement in dense haze regions while causing over-correction, color changes, or unnecessary alterations in low-haze regions. Additionally, Table 2 shows the effect of latent refinement depth on dehazing performance for both real and synthetic haze datasets. Compared with 4-step configuration, the 6-step latent refinement improves performance on the NH-Haze [1] dataset from 19.20 dB and 0.630 to 19.51 dB to 0.665 PSNR and SSIM, and Haze4K from 23.32 dB, 0.93 to 27.32 dB, 0.93 PSNR and SSIM. On O-Haze and synthetic haze datasets like SOTS Indoor, SOTS Outdoor, both 4-step and 6-step configurations achieve comparable results, indicating that, in contrast to relatively easier synthetic or less dense haze scenarios, additional refinement is more effective for dense, non-uniform, complex hazy scenes. For the final configuration of the proposed dehazing method, 6-step latent refinement is selected for the proposed model due to its superior performance on Nh-Haze [1],

SOTS Indoor [6], and Haze4K [7] while preserving identical 3.52 M parameter count. Wavelet guidance faithfully recovers high-frequency features such as edges, textures, and fine details; latent refinement enhances residual correction; adaptive gating regulates the refinement strength; and the dark channel provides haze-aware guidance, jointly enabling robust dehazing in dense and spatially non-uniform haze scenarios.

References

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